

Since the twig has travelled a shorter distance compared to the car, it will not hit the car.

Q2(a)

Power is work done per unit time or rate of doing work

2(b)(i)

$$P = Fv$$

$$= 2000 \times 45 = \mathbf{90 \text{ kW}}$$

2(b)(ii) 1. Work done = Power $\times$ time

$$= 90 \times 10^3 \times 3.0 \times 60 = 1.62 \times 10^7 \text{ J}$$

2(b)(ii)2.

$$\text{Increase in GPE} = mgh = 1200 \times 9.81 \times 3.3 \times 3.0 \times 60$$

$$= \mathbf{6.99 \times 10^6 \text{ J}}$$

2(b)(ii)3.

Work done against air resistance = Total work done – gain in GPE

$$= 1.62 \times 10^7 - 6.99 \times 10^6 = \mathbf{9.21 \times 10^6 \text{ J}}$$

2(b)(iii) Let air resistance be R .

$Rd = 9.21 \times 10^6$, where d = distance travelled by the plane in 3.0 min.

$$R = 9.21 \times 10^6 / (45 \times 3.0 \times 60) = \mathbf{1.14 \text{ kN}}$$

Q3

$$\text{Given } 20.2 = 1.66 \times 10^{27} \times 2.25 \times 10^{26} \times \frac{1}{d^2}$$